Projective involutions and support spaces in small dimension

Human Compression project: recovery of L011

This paper gives the geometric exclusions used in the simple-group classification of Leemans and Vandenschrick [1]. The dimension-three arguments rework Brooksbank and Vicinsky [2], including the projective qualification. The six-dimensional argument reworks Brooksbank and Leemans [3] for both quadratic types. Attribution to these arguments is retained. The deep rank-three existence classification is a separate input of the application.

A string marking is a sequence of distinct nonidentity involutions generating a group, with generators at distance at least two commuting. A string *C-group* marking also has the intersection property. Every subsequence of a C-group marking is independent: none of its generators lies in the subgroup generated by the others. We use this implication explicitly.

1 Projective dimension three

Let F be a finite field and $V = F^3$. A projective involution has a canonical linear involution representative. Indeed, if $X^2 = \mu I$, then $(\det X)^2 = \mu^3$, so μ is a square. In odd characteristic normalize X so its two-dimensional eigenspace has eigenvalue 1; the other eigenvalue is -1. In characteristic two normalize its unique eigenvalue to 1. Then $X = I + N$ with $N^2 = 0$ and $\operatorname{rank} N = 1$. In either characteristic the representative is uniquely determined by the projectivity and the specified normalization. Conjugation preserves it. Consequently commuting projectivities have commuting canonical representatives.

In odd characteristic write δ_X for the one-dimensional eigenspace and Δ_X for the other eigenspace. For distinct involutory projectivities, commutation is equivalent to

$$\delta_X \subset \Delta_Y, \qquad \delta_Y \subset \Delta_X. \tag{1}$$

This follows by preserving eigenspaces; their unequal dimensions prevent an exchange, and equal centers would force equal axes when the representatives commute. The word "distinct" matters: the criterion does not apply to an involution compared with itself. In characteristic two let $\delta_X = \operatorname{im} N$ and $\Delta_X = \ker N$. Here $\delta_X \subset \Delta_X$, and two transvections commute precisely when their centers coincide or their axes coincide. To check this, write $N = uf$, $M = vh$. If u, v are independent, $NM = MN$ forces both products to vanish, so both axes contain $\langle u, v\rangle$ and coincide. Equal centers conversely give zero products, as do equal axes.

Lemma 1 (Connectedness and absolute irreducibility)**.** *If a projective group on F^3 is irreducible and is generated by a string marking, then its canonical linear lift is absolutely irreducible and every adjacent pair is noncommuting.*

Proof. The lift is irreducible because invariant linear subspaces are the same as invariant projective subspaces. Its commuting endomorphism algebra is a finite division algebra. Any commuting endomorphism preserves the one-dimensional space δ_X of one of the canonical involutions. Restriction to that line is injective: a nonzero commuting endomorphism is invertible. The commuting algebra is therefore exactly F. The standard finite-dimensional algebra argument now gives absolute irreducibility: the faithfully acting algebra is semisimple (its radical annihilates a simple module), has one simple factor, and is a matrix algebra over its commuting division field, here F. Thus its image is the full endomorphism algebra and remains so after extending scalars.

If an adjacent pair commuted, the string would split into two nonempty parts whose generated linear groups commute. Over an algebraic closure an irreducible module of their direct product is a tensor product of irreducible modules. For completeness this is the matrix-algebra decomposition of the two commuting simple algebra images; each is the full matrix algebra on its tensor factor. The product of the factor dimensions is three, so one factor is one-dimensional. That group then acts by scalars, contrary to its containing a nonidentity projective generator. The same argument applies to any cut. □

Theorem 2 (Projective form obstruction)**.** *An irreducible string C-group in* $\mathrm{PGL}_3(F)$ *has rank at most four,* F *has odd characteristic, and its projective action lifts to a group preserving a nondegenerate symmetric bilinear form on* F^3*.*

Proof. Write the canonical lifts as $T_0, \dots, T_{r-1}$. Rank one is reducible. For rank two the two axes intersect in a nonzero invariant line. We may thus assume $r \geq 3$ and use Lemma 1.

First suppose the characteristic is two. The centers and axes of T_0, T_1 are distinct, since those involutions do not commute. If neither center lies in the other's axis, no transvection can commute with both: it would have to share the center or axis of each, which would force one of those incidences. Thus $r \leq 3$. The commuting endpoints of a triple share a center or an axis. In the former case their center together with the middle center spans an invariant plane; in the latter case their common axis intersects the middle axis in a common fixed line. In both cases the triple is reducible.

In the remaining case, duality permits us to assume $\delta_0 \subset \Delta_1$. The opposite incidence is impossible, since it would make both distinct axes the plane spanned by the centers. Every transvection commuting with both T_0, T_1 then has center δ_0 and axis Δ_1. These transvections commute with each other, so noncommuting adjacency gives $r \leq 4$. Also T_2 shares the center or axis of T_0 and hence fixes δ_0 pointwise. Every later generator has center δ_0 and fixes it, as do T_0, T_1. This contradicts irreducibility. Characteristic two is excluded.

Now suppose the characteristic is odd. If the first two centers coincide, all other generators preserve that center, by commuting with T_0 or T_1. If their axes coincide, the same argument gives a common plane. Hence both centers and axes are distinct. By (1), a projective involution commuting with both is uniquely determined, if it exists: its center is $\Delta_0 \cap \Delta_1$ and its axis is $\langle \delta_0, \delta_1 \rangle$. Two generators with indices at least three would therefore coincide. Independence gives $r \leq 4$.

For rank three, simultaneously diagonalize the commuting endpoints, giving their exceptional eigenspaces the first and third coordinate lines. The space of diagonal symmetric forms has dimension three. The symmetric forms preserved by T_1 have dimension four: in its eigenbasis they have one block of size one and one symmetric block of size two. Both spaces lie in the six-dimensional symmetric-matrix space, so their intersection is nonzero. Its radical is invariant under the full group and therefore zero.

For rank four, simultaneously diagonalize T_0, T_3 in the same way. The matrix T_1 preserves the coordinate plane $\langle e_1, e_2 \rangle$ and line $\langle e_3 \rangle$; T_2 preserves $\langle e_1 \rangle$ and $\langle e_2, e_3 \rangle$. Consider a diagonal symmetric form $B = \mathrm{diag}(x, y, z)$. For an involution T, invariance is equivalent to $BT = T^{\mathsf{T}}B$. The two block matrices impose at most one homogeneous equation each:

$$(T_1)_{12}x = (T_1)_{21}y, \qquad (T_2)_{23}y = (T_2)_{32}z.$$

There is a nonzero solution over F. Its radical is again invariant under all four generators, hence zero. This argument does not presume that the first rank-three subgroup is irreducible, or that the initial dihedral subgroup acts semisimply. □

Proposition 3. *No nonabelian simple group* $\mathrm{PSL}_3(q)$ *or* $\mathrm{PSU}_3(q)$ *has a string C-group marking of any rank.*

Proof. Their natural projective actions over $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$ are irreducible, a standard natural-module fact. Even characteristic is excluded by Theorem 2. For odd $q \geq 3$, a lifting group as in that theorem would lie in the orthogonal *isometry* group, whose order in dimension three over $\mathbb{F}_t$ is $2t(t^2-1)$. The usual classical-group orders give

$$\frac{|O_3(q)|}{|\operatorname{PSL}_3(q)|} = \frac{2\gcd(3,q-1)}{q^2(q^3-1)} < 1,$$

and

$$\frac{|O_3(q^2)|}{|\operatorname{PSU}_3(q)|} = \frac{2\gcd(3,q+1)(q^2+1)}{q(q^3+1)} \leq \frac{6(q^2+1)}{q(q^3+1)} < 1.$$

A group projecting onto the indicated simple group cannot have smaller order. The natural-module and order formulas are retained classical inputs; no finite-field sampling is used. □

2 Orthogonal dimension six in characteristic two

Let (V,Q) be a nonsingular six-dimensional quadratic space over the finite field F of characteristic two, of either type. Write B for its polar alternating form. We retain the classical Dickson-invariant identity

$$\Omega(V) = \{g \in O(V) : \operatorname{rank}(g-I) \text{ is even}\}. \tag{2}$$

Together with the two Klein identifications used below, this is the classical orthogonal-group input. The remaining support calculation is given explicitly.

Lemma 4 (Involution supports)**.** *For a nonidentity involution $g \in \Omega(V)$ its support $U = \operatorname{im}(g-I)$ has dimension two, is totally isotropic for B, and $\ker(g-I) = U^\perp$. If two such involutions commute, their supports intersect nontrivially. If they do not commute and their supports intersect, their intersection is a nonsingular line for Q.*

Proof. Put $N = g - I$. Since $N^2 = 0$, its rank is at most three; (2) makes it two. Because g is an involutory isometry, N is self-adjoint for B. It follows that $\ker N = U^\perp$ and that $B(Nx,Ny) = B(x,N^2y) = 0$. Choose a basis e,b of U with $Q(e) = 0$; such an e exists since the field is perfect and the polar form on U is zero. Self-adjointness and $Q(x+Nx) = Q(x)$ show that, for some $\alpha \neq 0$,

$$Nx = \alpha B(x,b)e + \alpha B(x,e)b + \alpha^2 Q(b)B(x,e)e. \tag{3}$$

Here is the coefficient verification. Write a self-adjoint map with image in U as $Nx = \alpha B(x,b)e + \alpha B(x,e)b + \beta B(x,e)e + \gamma B(x,b)b$. The quadratic identity gives $\beta = \alpha^2 Q(b)$ and $\gamma = \gamma^2 Q(b)$. If $\gamma \neq 0$, the coefficient matrix has determinant $\alpha^2(1+\gamma Q(b)) = 0$, contrary to rank two. Thus $\gamma = 0$, and rank two forces $\alpha \neq 0$.

For commuting N, M with disjoint images, $NM = MN$ has image in their intersection and is zero. Their two support planes are then mutually orthogonal, so their sum is a four-dimensional totally isotropic space for the nonsingular alternating form B on V, impossible. Thus their supports intersect. If the common line is singular, take its vector as e in (3) for both maps. Substitution gives $NM = MN$, both scalar multiples of the map $x \mapsto B(x,e)e$. Finally, equal support planes also give commuting maps, since each map kills that plane. Hence noncommuting supports meet, if at all, in a nonsingular line. □

Theorem 5 (The six-dimensional string obstruction)**.** *A sequence of involutions in $\Omega(V)$ with the string commutations and with noncommuting adjacent pairs generates a reducible group on V.*

Proof. Let $U_i = \operatorname{im}(g_i - I)$. The sum of all these supports is invariant under all generators, since $g_i(v) - v \in U_i$ for every v. For at most two generators its dimension is at most four. For three

it is at most five, because U_0 meets U_2. For four, both $U_0 + U_2$ and $U_1 + U_3$ have dimension at most three and they intersect, since U_0 meets U_3. Their sum is again proper.

Assume at least five generators. If U_0, U_1 are disjoint, each U_j for $j \geq 3$ meets both and lies in their four-dimensional sum. Since U_2 meets U_0, the total support has dimension at most five. Otherwise $U_0 \cap U_1 = \langle b \rangle$ with $Q(b) \neq 0$ by Lemma 4. Put $W = U_0 + U_1 \subset b^\perp$.

If U_1, U_2 are disjoint, put $W' = U_1 + U_2$. All U_j, $j \geq 4$, lie in W', and U_0 meets U_2. If $U_0 \subset W'$, the plane U_3 meets U_1 and adds at most one dimension. Otherwise $W' + U_0$ has dimension five. The plane U_3 meets both U_1 and U_0, so lies in $W' + U_0$ unless those two intersections are the same line. In the latter case that line lies in $U_1 \cap U_0$, which is disjoint from U_2; since $U_0 \cap W'$ already contains the line $U_0 \cap U_2$, two distinct lines would force $U_0 \subset W'$, a contradiction. Thus the total support is again proper.

We may therefore suppose $U_1 \cap U_2 = \langle c \rangle$ with $Q(c) \neq 0$. The plane U_2 also meets U_0. If it contains b, then $U_2 \subset b^\perp$; otherwise its two intersections span it and $U_2 \subset W \subset b^\perp$. Thus g_0, g_1, g_2 fix b. Each later U_j meets both U_0, U_1; it either contains b or lies in W. In both cases it is perpendicular to b, so g_j fixes b. The whole generated group has a fixed line. □

Proposition 6. *For every even prime power q, neither* $\mathrm{PSL}_4(q)$ *nor* $\mathrm{PSU}_4(q)$ *admits a string C-group marking of any rank.*

Proof. Retain the standard Klein identifications $\mathrm{PSL}_4(q) \cong \Omega_6^+(q)$ and $\mathrm{PSU}_4(q) \cong \Omega_6^-(q)$, with irreducible natural six-dimensional actions. In these even-characteristic parameters both are nonabelian simple. A commuting adjacent pair in a string C-group would split it as a direct product of two nontrivial subgroups, contrary to simplicity. Theorem 5 now excludes a generating marking. This proves the minus-type transfer explicitly, including $q = 2$; it does not identify the simple derived group with the full orthogonal group. □

References

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[3] P. A. Brooksbank and D. Leemans, *Polytopes of large rank for* $\mathrm{PSL}(4, q)$, J. Algebra **452** (2016), 390–400. The 26 November 2015 author manuscript, Section 4, was read in the author's ResearchGate full-text representation. It is the fixed-catalogue L048 cited as a proof prerequisite, not a new recovery or source count.

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